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REVIEW

# Genetics and Primary Aldosteronism: Disease Phenotype and Demographic Implications for Hypertension

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**ABSTRACT:** Detecting secondary forms of hypertension is key to targeted management and prevention of cardiovascular complications. Primary aldosteronism (PA) is the most common form of secondary arterial hypertension, resulting from excess adrenal unilateral or bilateral aldosterone production. Its prevalence rises with hypertension severity, reaching 25% in treatment-resistant patients. Understanding the genetic basis of PA provides key insights into its pathophysiology and improves strategies for diagnosing and managing hypertension in the general population. Genetic studies performed over the last few years have unraveled the genetic spectrum of PA. Somatic and germline mutations have been identified in most aldosterone-producing adenomas and familial forms of the disease. Recent studies have also uncovered genetic loci that increase susceptibility to PA that are common to lateralized and bilateral forms of PA and shared with blood pressure loci. Together with the clinical and biochemical description of a continuum of aldosterone dysregulation in hypertension, these findings mark a shift in our understanding of disease pathogenesis, indicating that common genetic variants may contribute to dysregulated aldosterone production in hypertension and to the development of PA in extreme cases. Different genetic susceptibility and genotype-phenotype correlations may shape the demographic distribution and clinical recognition of PA, influencing both its apparent prevalence and the likelihood of timely diagnosis.

**Key Words:** adrenal glands ■ aldosterone ■ genetics ■ hyperaldosteronism ■ hypertension



Arterial hypertension (HT) accounts for >10 million deaths/y from stroke, heart disease, kidney disease, and other vascular conditions.[1] Despite available treatments, up to two-thirds of patients fail to achieve optimal blood pressure control.[2] Detecting secondary forms of HT is key to targeted management and prevention of cardiovascular complications. Primary aldosteronism (PA) is the most common form of secondary HT, resulting from excess adrenal aldosterone production that is, in the majority of cases, either unilateral, from an aldosterone-producing adenoma (APA), or due to bilateral adrenal hyperplasia (BAH, also called idiopathic hyperaldosteronism).[3] Its prevalence rises with HT severity, reaching 25% in treatment-resistant HT.[4] PA remains largely underdiagnosed due to its complex workup, contributing to significant cardiovascular burden.[5–8] Beyond overt PA, recent evidence indicates a continuum of renin-independent dysregulated aldosterone production affecting a substantial proportion of patients with HT in the general population.[9]

PA, also known as Conn syndrome, was first described by Jerome W. Conn in 1955 in a patient presenting with muscle weakness along with intermittent muscle spasms. These symptoms were accompanied by hypertension resistant to treatment as well as polydipsia, polyuria, hypokalemia, proteinuria, and high aldosterone levels. These symptoms were found to be associated with an APA.[10,11] PA has since been identified as a frequent cause of HT, accounting for 5% of subjects with hypertension in primary care and up to 20% in patients with treatment-resistant hypertension.[4,12] It is particularly frequent in patients under 40 years referred to specialized centers, where it accounts for 16% of cases.[13] PA is characterized by high plasma aldosterone levels, renin suppression, associated with an elevated aldosterone-to-renin ratio and hypokalemia in 9% to 37% of cases.[14] Elevated

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## Nonstandard Abbreviations and Acronyms

| | |
|---|---|
| **Ang II** | angiotensin II |
| **APA** | aldosterone-producing adenoma |
| **APCC** | aldosterone-producing cell cluster |
| **APM** | aldosterone-producing micronodule |
| **AS** | aldosterone synthase |
| **BAH** | bilateral adrenal hyperplasia |
| **GPCR** | G protein–coupled receptor |
| **HT** | arterial hypertension |
| **PA** | primary aldosteronism |

aldosterone increases cardiovascular risk beyond that of essential hypertension due to its damaging effects on the heart and vessels, independent of blood pressure levels.[6] This is due to the direct effects of aldosterone in various tissues that drive end-organ damage and cardiovascular disease.[15] Given the high prevalence of PA and the severe cardiovascular consequences of aldosterone excess, the 2025 Endocrine Society guidelines suggest screening of all patients with hypertension for PA by measuring aldosterone and renin levels.[5]

Somatic and germline mutations have been identified in the majority of APA and in familial forms of PA.[16] More recently, genetic loci have been identified that confer an increased risk of developing PA.[17,18] This represents a paradigm shift in our understanding of the pathogenesis of the disease, suggesting that common genetic variants may lead to dysregulated aldosterone production in hypertension and the development of PA in extreme cases. This review presents our current knowledge on the genetics of PA, discusses pathogenic mechanisms and genotype-phenotype correlations, variability across sex, age, and population structure, and their potential implications for stratifying PA in the general population with hypertension.



## ALDOSTERONE PRODUCTION IN THE ADRENAL CORTEX

### Adrenal Cortex Structure and Cell Lineage Conversion

The adrenal cortex is organized into 3 anatomically and functionally distinct zones: the zona glomerulosa (ZG), the zona fasciculata (ZF), and the zona reticularis. Thanks to a peculiar functional zonation, each of these zones is responsible for the secretion of specific steroid hormones. Mineralocorticoid hormones, primarily aldosterone, are synthesized in the ZG, which constitutes the outer layer of the adrenal cortex, accounting for 5% to 10% of its size. Glucocorticoid hormones, mainly cortisol in humans and corticosterone in mice and rats, are produced in the ZF. Adrenal androgens, specifically dehydroepiandrosterone, are synthesized in the zona reticularis.[19]

The adrenal cortex is a highly dynamic organ, undergoing continuous renewal throughout life. Several studies in mice have allowed to determine adrenal cortex development, renewal, and lineage conversion. Stem/progenitor cells located in the adrenal capsule and subcapsular ZG contribute to adult adrenal cortex cell turnover. These cells proliferate and migrate centripetally, differentiating into ZG and subsequently into ZF cells, before undergoing apoptosis at the corticomedullary junction.[20] This centripetal model of zonation has been supported by different lineage tracing studies showing radial migration of *Shh* and *Gli1*-expressing progenitor cells.[21]

Transcriptional regulators play a crucial role in cell fate determination. SF1 (steroidogenic factor 1/*NR5A1*) is essential for adrenal development and steroidogenic gene expression. Sf1 knockout mice lack adrenal glands entirely, and heterozygous mice show impaired compensatory growth after unilateral adrenalectomy.[22,23] SF1 also facilitates lineage conversion, enabling mesenchymal stem cells to differentiate into steroid-producing cells. Recent lineage tracing studies demonstrated that ZG cells can convert into ZF cells postnatally[24,25] and contribute in a major way to adrenal cortex renewal in adults.

Homeostatic renewal of the cortex is regulated by RSPO (R-spondin)/WNT/β-catenin and SHH (Sonic hedgehog protein) signaling pathways. RSPO3 from capsular cells activates Wnt signaling in subcapsular progenitors, which express *SHH* and *WNT4*, promoting their differentiation into steroidogenic cells. As proliferation occurs mostly in the capsule and outer cortex, newly generated cells undergo differentiation into ZG cells, which then undergo centripetal migration and lineage conversion into ZF cells.[20] These processes are regulated by the mutually antagonistic WNT/β-catenin and cAMP-PKA (Protein Kinase A) signaling pathways, with WNT signaling promoting ZG identity and cAMP-PKA signaling promoting ZF differentiation.[26,27]

### Regulation of Aldosterone Biosynthesis

Adrenal steroids are synthesized from cholesterol by a series of enzymatic steps involving CYP (cytochrome) P450 family enzymes and hydroxysteroid dehydrogenases. These reactions take place either in the mitochondrial membrane or in the endoplasmic reticulum. The production of zone-specific steroids results from differential enzyme expression: the ZG expresses *CYP11B2* to produce mineralocorticoids, whereas the ZF and zona reticularis express *CYP11B1* and *CYP17A1* for glucocorticoid and androgen biosynthesis, respectively.[28]

In the ZG, AS (aldosterone synthase), encoded by the *CYP11B2* gene, catalyzes the final steps of aldosterone

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production.[29] The biosynthetic pathway begins with the transport of cholesterol to the inner mitochondrial membrane by the StAR (steroidogenic acute regulatory protein). There, *CYP11A1* catalyzes the cleavage of the cholesterol side chain to form pregnenolone, which is then transferred to the endoplasmic reticulum. Here, 3β-HSD (3β-hydroxysteroid dehydrogenase) converts pregnenolone into progesterone, which is hydroxylated by 21-hydroxylase (*CYP21A2*) to produce 11-deoxycorticosterone. This intermediate returns to the mitochondria, where aldosterone synthase catalyzes 3 sequential reactions: the 11β-hydroxylation of 11-deoxycorticosterone to form corticosterone, followed by 18-hydroxylation of corticosterone to produce 18-hydroxycorticosterone, and finally, an 18-oxidation step converting 18-hydroxycorticosterone into aldosterone.[29] Aldosterone plays a key role in regulating blood pressure and electrolyte balance by promoting sodium reabsorption and potassium excretion in the distal nephron of the kidney via its binding to the mineralocorticoid receptor. Aldosterone also exerts effects on other tissues such as the colon, salivary and sweat glands, heart, brain, and adipose tissue. Inappropriate mineralocorticoid activation in extrarenal target tissues is associated with cardiac hypertrophy, fibrosis, vascular remodeling, insulin resistance, and inflammation in adipose tissue.[30]

Aldosterone production is primarily regulated by the renin-angiotensin system and extracellular potassium levels, as well as adrenocorticotropic hormone (ACTH).[29] Ang II (angiotensin II) binds to the AT1 receptor on ZG cells, a GPCR (G-protein–coupled receptor) linked to Gαq, activating phospholipase C and generating inositol 1,4,5-trisphosphate and diacylglycerol. Inositol 1,4,5-trisphosphate stimulates calcium release from the endoplasmic reticulum, increasing intracellular calcium and activating CaMKs (calcium/calmodulin-dependent protein kinases), which phosphorylate transcription factors, such as NR4A1 (Nuclear receptor subfamily 4 group A member 1, *NUR77*), NR4A2 (Nuclear receptor subfamily 4 group A member 2, *NURR1*), and CREBH (cAMP-responsive element–binding protein H), enhancing *CYP11B2* expression. Ang II also inhibits potassium channels, such as TASK1 (Potassium channel subfamily K member 3), TASK3 (Potassium channel subfamily K member 9), and GIRK4 (G-protein–activated inward rectifier potassium channel 4; *KCNJ5*), and the $Na^+$/$K^+$-ATPase, leading to membrane depolarization and opening of voltage-gated calcium channels. ZG cells are highly sensitive to extracellular potassium due to their high expression of potassium channels. Elevated $K^+$ levels reduce potassium conductance, depolarize the membrane, and activate calcium signaling.[31] Beyond these primary regulators, ACTH can acutely stimulate aldosterone synthesis via the MC2R (Melanocortin receptor 2), a GPCR linked to Gαs, which activates adenylate cyclase and the PKA pathway. PKA phosphorylates StAR directly or indirectly via CREB (cAMP-responsive element–binding protein), increasing cholesterol transport to mitochondria. ACTH also upregulates *CYP11A1*, enhancing precursor availability.[32] In healthy individuals, ACTH stimulation produces substantial increases in plasma aldosterone concentrations.[33] Chronic ACTH exposure, however, suppresses aldosterone production and promotes transdifferentiation of ZG cells into ZF-like cells.[34]

## HISTOLOGY OF ALDOSTERONE BIOSYNTHESIS IN THE ADRENAL CORTEX AND EVOLUTION WITH AGE

Analysis of *CYP11B2* or AS expression in the adrenal cortex from adult subjects has shown discontinuous ZG and the presence of focal aldosterone-producing areas, originally called aldosterone-producing cell clusters (APCC),[35,36] recently renamed aldosterone-producing micronodules (APMs).[37] APCC/APMs comprise subcapsular ZG-like cells overlying inner, larger ZF-like cells and selectively express *CYP11B2* but not *CYP11B1*, consistent with an aldosterone-producing phenotype.[35,38] Transcriptomic profiling showed that APCC/APMs are molecularly close to ZG with enhanced aldosterone capacity, supporting the concept that they are foci of constitutive aldosterone biosynthesis.[39] The adrenal ZG undergoes major histological changes with age, with accumulation of APCC/APMs to replace the continuous ZG observed in young adrenals.[40] Remarkably, somatic mutations have been identified in APCC/APMs from subjects with no evidence of PA,[40] which are supposed to be associated with dysregulated aldosterone production.[41] The relation of autonomous aldosterone production and age has been investigated by evaluating renin activity and aldosterone in subjects on different salt diets.[41] Aging was associated with a decline in renin activity under a high-salt diet, whereas aldosterone levels remained relatively stable, raising the aldosterone-to-renin ratio. This was associated with a blunted aldosterone response under sodium restriction, consistent with an age-related shift towards autonomous aldosterone production.[41] Similarly, in another cohort of subjects without PA, the total CYP11B2-positive area in the ZG decreased as the number of APCC/APMs increased, suggesting a role for APCC/APMs in the pathophysiological autonomous aldosterone production in older subjects.[41]

In addition to somatic mutational processes, emerging evidence indicates that epigenetic mechanisms, including DNA methylation, histone modifications, and noncoding RNA–mediated regulation, also contribute to dysregulated aldosterone production and adrenal cellular remodeling.[42] APAs display a distinct hypomethylated epigenetic landscape, particularly within the *CYP11B2* promoter and across networks of steroidogenic, GPCR-linked, and tumor-related genes, creating a transcriptionally permissive state that enhances aldosterone biosynthesis.[42–44] Age-related methylation drift within the adrenal cortex, together with the accumulation of APCC/APMs, suggests that epigenetic dysregulation may be a complementary determinant of age-related autonomous aldosterone production.

## SOMATIC GENETICS OF APA, APCC/APMS, ALDOSTERONE-PRODUCING NODULES AND BAH

Recurrent somatic mutations in different genes have been identified in APA.[16] The use of CYP11B2 (AS) immunohistochemistry-guided next-generation sequencing for genetic screening has enabled precise detection of somatic mutations, revealing that ≈90% of APA carry mutations in genes involved in ionic transport and regulation of membrane potential of ZG cells, including *KCNJ5, CACNA1D, ATP1A1*, *ATP2B3, CLCN2, CACNA1H, SLC30A1*, and *MCOLN3*.[16,45,46] Mutations in other genes, such as *CTNNB1*, *GNAQ/GNA11*, and *CADM1*, revealed additional molecular mechanisms leading to APA development.[47,48] In addition, immunohistochemistry for AS has identified several histological subtypes of lesions in PA lateralized at adrenal vein sampling, including APA, aldosterone-producing nodules, APCC/APMs, as well as diffuse aldosterone-producing hyperplasia.[37] Often, different histological abnormalities coexist in the same adrenal gland, with different aldosterone-producing structures carrying different somatic mutations.[49,50] Nevertheless, aldosterone-producing structures in adrenals with APA share common molecular characteristics and cellular environment, as revealed by immunohistological characterization of different key molecules of adrenocortical homeostasis, suggesting common developmental mechanisms.[51]

Somatic mutations in APA-driver genes have been identified in APCC/APMs within adrenals with APA. Although in subjects without hypertension, *CACNA1D* mutations are most frequently detected in APCC/APMs, and *KCNJ5* mutations have not been observed,[39] in APCC/APMs located within adrenals with APA, a different mutational pattern has been described, including APCC/APMs carrying *KCNJ5* mutations.[50] Remarkably, genetic heterogeneity has been observed between the APA itself, adjacent AS-positive secondary nodules, and independent APCC/APMs within the same adrenal. This diversity of mutational profiles among aldosterone-producing structures in a single adrenal supports the existence of complex and multilayered mechanisms driving APA development. In this context, spatial and single-cell/nucleus transcriptomic analysis, exploring evolutionary trajectories of APA development from APCC/APMs, suggest that APA with *KCNJ5* mutations develop directly from ZG, while APA with other mutations may derive from APCC/APMs through a stepwise trajectory.[52] Finally, accumulation of APCC/APMs that carry somatic mutations similar to APA has also been observed in adrenals from patients with BAH, suggesting a similar pathogenesis between the lateralized and bilateral forms of PA.[53]

## PATHOGENIC MECHANISMS OF ALDOSTERONE PRODUCTION IN APA AND GENOTYPE-PHENOTYPE CORRELATIONS

Genetic abnormalities identified in APA mainly affect proteins regulating intracellular ion balance and membrane potential in ZG cells (Figure 1). They increase intracellular calcium signaling, the central pathway driving aldosterone production. Mutations in *KCNJ5*, *CLCN2*, *ATP1A1*, and *SLC30A1* promote cell membrane depolarization by altering ion conductance and disrupting the electrophysiological identity of ZG cells. *KCNJ5* variants impair potassium selectivity of GIRK4 channels, promoting inappropriate $Na^+$ influx and membrane depolarization.[54] Gain-of-function *CLCN2* mutations cause constitutive chloride efflux in ZG cells, also leading to cell membrane depolarization.[55–57] Mutations in *ATP1A1* impair $Na^+/K^+$-ATPase activity, generating inward leak of sodium and hydrogen ions,[58,59] while mutations in *SLC30A1,* coding for the zinc transporter ZnT1 (Zinc transporter 1), similarly induce pathological $Na^+$ influx and cell membrane depolarization.[45] All these genetic defects shift the ZG cell membrane potential towards depolarized states, increasing the opening probability of voltage-gated calcium channels and thereby raising intracellular $Ca^{2+}$ concentrations.

A second major group of pathogenic variants acts by directly increasing intracellular calcium concentrations (Figure 1). Activating mutations in *CACNA1D* and *CACNA1H,* coding for the calcium channels Cav1.3 (Voltage-dependent L-type calcium channel subunit alpha-1D) and Cav3.2 (Voltage-dependent T-type calcium channel subunit alpha-1H), shift their voltage-dependence towards more negative membrane potentials, enabling these channels to open under near-physiological conditions and thereby sustaining tonic $Ca^{2+}$ influx.[59–62] Mutations in *ATP2B3*, which encodes PMCA3 (Plasma membrane calcium-transporting ATPase 3), reduce the efficiency of $Ca^{2+}$ extrusion, prolonging cytosolic $Ca^{2+}$ elevations.[63] Mutations in *MCOLN3*, encoding the mucolipin family cation-permeable channel TRPML3 (Transient receptor potential channel mucolipin 3), may contribute through a dual mechanism: increasing nonselective cation permeability that promotes depolarization while simultaneously enhancing $Ca^{2+}$ entry through TRP-mediated currents.[46] All together, these alterations produce a persistent $Ca^{2+}$ signal that activates $Ca^{2+}$-dependent kinases and transcription factors, ultimately driving increased *CYP11B2* expression and autonomous aldosterone biosynthesis.

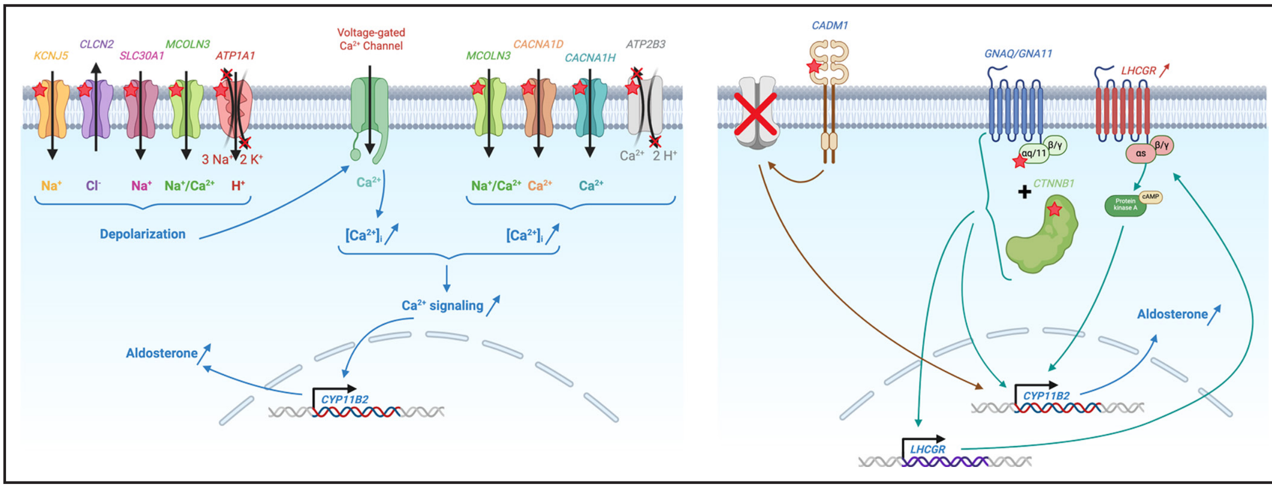


**Figure 1. Pathogenic mechanisms of autonomous aldosterone production in primary aldosteronism (PA).**
Genetic alterations in PA converge on two principal mechanisms that increase intracellular calcium concentration. One major pathway involves mutations that depolarize the zona glomerulosa cell membrane: *KCNJ5* variants impair $K^+$ selectivity and allow $Na^+$ influx; *CLCN2* mutations cause constitutive $Cl^-$ efflux; *ATP1A1* mutations reduce $Na^+/K^+$-ATPase activity and generate inward $Na^+/H^+$ leak currents; and *SLC30A1* mutations promote pathological $Na^+$ entry. These defects shift the cell membrane potential toward depolarized states and increase voltage-gated $Ca^{2+}$ channel opening, thereby elevating intracellular $Ca^{2+}$ concentrations. A second mechanism comprises mutations that directly elevate intracellular $Ca^{2+}$. Gain-of-function *CACNA1D* and *CACNA1H* variants enhance Cav1.3 (Voltage-dependent L-type calcium channel subunit alpha-1D) and Cav3.2 (Voltage-dependent T-type calcium channel subunit alpha-1H) activation at more negative potentials, while *ATP2B3* mutations impair PMCA3 (Plasma membrane calcium-transporting ATPase 3)-mediated $Ca^{2+}$ extrusion. *MCOLN3* mutations exert a dual effect by increasing non-selective cation permeability, promoting depolarization, and facilitating $Ca^{2+}$ influx through TRP-mediated currents. Together, these alterations generate a sustained $Ca^{2+}$ signal that upregulates *CYP11B2* expression and drives autonomous aldosterone production. Additional pathogenic mechanisms contribute to autonomous aldosterone production. Activating *CTNNB1* mutations generate constitutive WNT/β-catenin signaling, stimulating aldosterone biosynthesis and adrenal cell proliferation. Some aldosterone-producing adenomas carry combined *CTNNB1* and activating *GNAQ* or *GNA11* variants, which amplify Gαq/11 (Guanine nucleotide-binding protein G(q) subunit alpha/ Guanine nucleotide-binding protein G(11) subunit alpha) signaling and drive LHCGR (Luteinizing hormone/choriogonadotropin receptor) overexpression, linking adenoma formation to periods of increased luteinizing hormone/hCG (human chorionic gonadotropin) activity, such as puberty, pregnancy, and menopause. Finally, *CADM1* alterations disrupt gap junction communication in the zona glomerulosa, removing inhibitory control of *CYP11B2* and facilitating autonomous aldosterone secretion.

Beyond defects in ion homeostasis and $Ca^{2+}$-handling, additional mechanisms have been recently discovered that lead to autonomous aldosterone production in APA (Figure 1). Activating mutations in *CTNNB1* lead to constitutive WNT/β-catenin signaling, promoting both aldosterone biosynthesis and proliferative activity within adrenal cells.[64] In some APA, occurring at pregnancy, puberty, and menopause, *CTNNB1* mutations co-occur with activating mutations in *GNAQ* or *GNA11*, driving Gαq/11 (Guanine nucleotide-binding protein G(q) subunit alpha/Guanine nucleotide-binding protein G(11) subunit alpha)-mediated signaling and increasing *LHCGR* expression. This mechanistic interplay links APA development and hormonally dynamic physiological states such as puberty, pregnancy, or menopause, when luteinizing hormone/hCG (human chorionic gonadotropin) pathway activity is elevated.[47] Finally, recent evidence implicates *CADM1*, coding for the membrane-adhesion and cell-organization cellular adhesion molecule 1, in specific APA subtypes. Altered CADM1 function has been reported to disrupt gap junction–mediated intercellular communication in ZG cells, removing a physiological suppressive signal to *CYP11B2* expression and leading to autonomous aldosterone production.[48] Finally, aberrant expression and activation of multiple G-protein–coupled receptors and their ligands may contribute to or potentiate the dysregulated aldosterone production in PA.[65]

Different studies have explored the genotype-phenotype correlations of APA-driver mutations. Somatic *KCNJ5* mutations are significantly more frequent in women and younger patients with APA.[66–68] Several investigations also report larger tumor size and a more florid biochemical phenotype, including higher preoperative aldosterone concentrations. Although this association is not uniformly observed across all studies and may depend on the study population.[67,68] Clinically, *KCNJ5* mutations have emerged as a potential prognostic marker. Patients harboring *KCNJ5* mutations showed higher rates of complete clinical and biochemical success after adrenalectomy and greater postoperative improvements in cardiac parameters compared with non-mutant cases.[69,70] A strong genotype–phenotype association has also been described for patients harboring double *CTNNB1* and *GNAQ* or *GNA11* mutations, who frequently present with APA during puberty, pregnancy, or menopause, consistent with the mechanisms outlined above.[47] Collectively, these findings delineate specific molecular subgroups of APA, improving genetic diagnosis and enabling more refined disease detection. Age-related differences were also described. In young patients, lateralized PA is dominated by APA with *KCNJ5* mutations, while *CACNA1D* mutations were more frequent in aldosterone-producing nodules.[71] In older patients, adrenal morphology becomes more heterogeneous, with aldosterone-producing nodules more common than APA and *CACNA1D* mutations increasingly represented.[72]

In adrenals with APA, pronounced cellular heterogeneity has been demonstrated. *KCNJ5*-mutated APA typically show a predominance of clear cells, heterogeneous *CYP11B2* and *CYP11B1* expression, and a specific steroid profile characterized by elevated hybrid steroids 18-hydroxycortisol and 18-oxocortisol, reflecting their unique steroidogenic activity.[50] In addition, *KCNJ5*-mutated APA exhibit reduced GIRK4 protein expression relative to adjacent ZG, despite similar mRNA levels, suggesting channel dysfunction potentially contributing to membrane depolarization and aldosterone excess.[14] Additional layers of adrenal histological heterogeneity arise from region-specific activation of Wnt/β-catenin and ACTH signaling, dense mast-cell infiltration, and paracrine regulatory influences within *CYP11B2*-positive zones of APA and APCC/APMs, showing that the local microenvironment collectively shapes zonal identity and steroid output.[51,73] These features align with findings from single-cell, spatial transcriptomics, and metabolomics, which reveal pronounced genotype-dependent transcriptional and metabolic heterogeneity.[74] Distinct *CYP11B2*-positive cell signatures were linked to *KCNJ5* mutation status: oxidative stress–associated metabolites accumulate in wild-type tumors, whereas antioxidant profiles characterize *KCNJ5*-mutant lesions.[74] In addition, intratumoral heterogeneity within *KCNJ5*-mutated APA is characterized by distinct cell subpopulations,[75] with distinct cellular trajectories characterized by the activation of either the ribosome and neurodegenerative disease pathways or the glycolysis pathway.[75] In a different multiomics analysis, cellular heterogeneity in APA with *KCNJ5* mutations was associated with tumor cell reprogramming progressing from stress-responsive cells to aldosterone-producing or cortisol-producing cells, which were spatially separated within the tumor.[76] Lipid-associated macrophages were identified as potential drivers of cortisol-producing cell subpopulations, linking intratumor cortisol synthesis to increased circulating cortisol levels and complications in PA, suggesting that the tumor microenvironment may contribute to phenotypic diversity and clinical outcomes associated with different APA genotypes.[76]

## FAMILIAL FORMS OF PA

Familial hyperaldosteronism (FH) is a rare disease and accounts for ≈1% to 5% of PA cases.[77] Different germline mutations in 4 genes define 4 Mendelian forms of FH (I–IV), which are inherited in an autosomal dominant pattern.

FH type I (FH-I, OMIM no. 103900), also known as glucocorticoid-remediable aldosteronism, is a rare subtype of FH caused by a chimeric *CYP11B1/CYP11B2* gene resulting from unequal crossing-over between *CYP11B1*, which encodes 11β-hydroxylase responsible for cortisol biosynthesis, and *CYP11B2*, which encodes AS.[78,79] This genetic rearrangement results in the fusion of the promoter sequence of *CYP11B1* with the coding sequence of *CYP11B2*, leading to the expression of AS under the control of the *CYP11B1*, making aldosterone biosynthesis dependent on the ACTH rather than regulated by the renin-angiotensin system.[78] Consequently, aldosterone is produced ectopically in the ZF instead of the ZG, after the circadian rhythmicity of cortisol, leading to excessive aldosterone production and subsequent hypertension, with suppressed plasma renin activity and elevated aldosterone levels. Clinically, patients often exhibit elevated levels of hybrid steroids, such as 18-hydroxycortisol and 18-oxocortisol, which are considered diagnostic markers of FH-I. A defining feature of FH-I is its suppressibility by exogenous glucocorticoids. Administration of low-dose dexamethasone suppresses ACTH secretion, thereby reducing aldosterone synthesis and normalizing blood pressure and potassium levels[3,80,81]

FH types II to IV are due to germline pathogenic variants in genes also involved in APA (Figure 1). FH type II (OMIM no. 605635) is a form of PA that is not responsive to glucocorticoid therapy. It was initially mapped to the locus 7p22 through linkage analysis in several affected families.[82,83] The condition is inherited in an autosomal dominant pattern and presents with variable severity, ranging from mild hypertension to early onset, treatment-resistant forms. The molecular basis of FH type II was clarified with the identification of heterozygous mutations in the *CLCN2* gene (3q27.1), which encodes the ClC-2 chloride channel, in a large Australian family,[55] with another mutation described in parallel in a case of early onset PA.[56] Mutations are typically located within a short amino acid segment of ClC-2 called the inactivation domain, resulting in constitutive activation and loss of normal regulatory responses to voltage, pH, or cell volume. The persistent chloride efflux leads to membrane depolarization, activation of voltage-gated calcium channels, and increased *CYP11B2* transcription, thereby stimulating aldosterone production.[56] A certain degree of phenotypic variability and incomplete penetrance observed in FH type II is thought to reflect the diversity of *CLCN2* mutations, which can affect different functional domains of the channel.[55,84]

FH type III (OMIM no. 613677) is a rare form of PA caused by germline mutations in the *KCNJ5* gene. Like mutations responsible for APA, these mutations disrupt the ion selectivity filter of the channel, allowing sodium to enter ZG cells. This abnormal influx leads to membrane depolarization, activation of voltage-gated calcium channels, and an increase in intracellular calcium levels. The elevated calcium concentration stimulates the expression of *CYP11B2*, resulting in excessive aldosterone production.[54,85] Clinically, FH type III typically presents with early onset PA with treatment-resistant hypertension, often during childhood or adolescence, and is frequently accompanied by hypokalemia, polyuria, and polydipsia.[86] Adrenal imaging may reveal diffuse or nodular bilateral adrenocortical hyperplasia. Biochemical findings include elevated plasma aldosterone, suppressed renin activity, and increased levels of hybrid steroids, such as 18-oxocortisol and 18-hydroxycortisol.[86,87] Two clinical forms of FH type III have been described. In severe cases, patients develop drug-resistant hypertension with significant hypokalemia at a young age, due to BAH, often requiring bilateral adrenalectomy to cure hypertension, followed by lifelong hormone replacement therapy.[54] In contrast, milder forms of the disease may be managed effectively with mineralocorticoid receptor antagonists, such as spironolactone or eplerenone.[87–89] A genotype-phenotype correlation has been observed in *KCNJ5* mutations, with variants such as p.T158A, p.I157S, p.E145Q, and p.G151R being associated with an early onset and severe forms of PA,[54] whereas p.G151E and p.Y152C are generally linked to milder clinical presentations.[88,90]

FH type IV (OMIM no. 617027) is a rare form of PA caused by heterozygous mutations in the *CACNA1H* gene, which encodes the α1 subunit of the voltage-gated Cav3.2 T-type calcium channel. These mutations were first identified in 2 independent studies, which together described 5 distinct variants in patients with different clinical presentations of PA.[61,62] Electrophysiological characterization of these variants revealed a gain-of-function phenotype, with altered channel kinetics that include activation at more hyperpolarized membrane potentials and delayed inactivation. These changes result directly in enhanced calcium influx, which promotes sustained intracellular calcium signaling. This, in turn, stimulates the expression of *CYP11B2*, leading to increased aldosterone production. Functional studies using H295R-S2 adrenocortical cells expressing the mutant Cav3.2 channels confirmed that these mutations enhance aldosterone biosynthesis, especially in response to potassium stimulation.[62] The clinical phenotype associated with FH type IV is variable. Some individuals present with early-onset hypertension and neurodevelopmental abnormalities, while others exhibit features resembling FH type II. This phenotypic diversity suggests that *CACNA1H* mutations may contribute to a spectrum of aldosterone-related disorders, depending on the specific mutation and its functional impact.[62]

A rare and complex clinical phenotype characterized by severe early onset PA associated with seizures and neurological abnormalities has been described in children harboring de novo heterozygous germline mutations in the *CACNA1D* gene, which encodes the α1 subunit of the voltage-gated Cav1.3 L-type calcium channel.[60] These mutations confer a gain-of-function effect, directly increasing calcium influx and intracellular calcium signaling, upregulating *CYP11B2* expression. This syndrome is referred to as PA, seizures, and neurological abnormalities (OMIM no. 615474). Congenital hypoglycemia has been observed in 2 affected individuals.[91,92] The Table summarizes the genetic, molecular, and clinical characteristics of FH.

## GENETIC SUSCEPTIBILITY TO PA AND ALDOSTERONE DYSREGULATION IN HYPERTENSION

Recent clinical, biochemical, and histological evidence has suggested a continuum of dysregulated aldosterone production in hypertension and common pathogenic mechanisms between unilateral and bilateral forms of PA. A continuum of renin-independent aldosterone dysregulation has been observed in subjects across all blood pressure categories, from normotension to hypertension, after a sodium suppression test.[9,93] Furthermore, adrenals from healthy individuals show APCC/APMs carrying somatic mutations in genes involved in PA. The number of APCC/APMs increases with age and is paralleled by an increase in autonomous aldosterone production and dysregulated aldosterone response to physiological stimuli in older individuals.[39–41] Accumulation of APMs carrying somatic mutations similar to APA has also been observed in adrenals from patients with BAH.[53] Finally, ≈6% of patients undergoing adrenalectomy for lateralized aldosterone production do not show complete biochemical cure after surgery, identifying patients with BAH with asymmetrical aldosterone production rather than unilateral APA.[94] Nevertheless, adrenals from these patients show bona fide APA carrying somatic mutations in PA driver genes,[94,95] supporting a common origin of lateralized and bilateral forms of PA.

Insight on possible mechanisms underlying these observations came recently from genome-wide association studies, which identified genetic risk loci for PA.[17,18] By applying a genome-wide association studies to a large cohort of patients, major genetic risk loci were identified on chromosomes 1, 13, and X.[17] Two candidate genes within the identified regions, castor zinc finger 1 (*CASZ1*) on chromosome 1 and relaxin family peptide receptor 2 (*RXFP2*) on chromosome 13, are expressed in human and mouse adrenal cortex and APA.[17] Functional studies in adrenocortical cells showed that overexpression of CASZ1 and RXFP2 significantly suppressed mineralocorticoid output under basal and stimulated conditions. Remarkably, PA loci identified in genome-wide association studies have been associated with HT-related traits

**Table. Genetic, Molecular, and Clinical Characteristics of FH Types**

| FH type | Gene | Protein | Molecular effect | Clinical presentation |
|---|---|---|---|---|
| FH-I (GRA; OMIM no.103900) | Chimeric *CYP11B1/ CYP11B2* | Steroid 11-β-hydroxylase (cytochrome P450 11B1)/ aldosterone synthase (cytochrome P450 11B2) | Aldosterone synthase stimulation by ACTH | Early-onset PA, increased 18-hydroxycortisol, and 18-oxocortisol. Suppressible by glucocorticoids |
| FH-II (OMIM no. 605635) | *CLCN2* | Chloride channel protein 2 (ClC-2) | Increased chloride efflux | Early-onset PA with phenotypic variability and incomplete penetrance |
| FH-III (OMIM no. 613677) | *KCNJ5* | G-protein–activated inward rectifier potassium channel 4 (GIRK4) | Altered potassium selectivity and increased sodium influx | Severe early-onset PA, treatment-resistant hypertension, severe hypokalemia, increased 18-hydroxycortisol and 18-oxocortisol, and bilateral adrenocortical hyperplasia; polyuria and polydipsia in some patients. |
| FH-IV (OMIM no. 617027) | *CACNA1H* | Voltage-gated T-type calcium channel subunit $\alpha$-1H (Cav3.2) | Increased calcium influx | Early-onset PA with phenotypic variability and incomplete penetrance; neurodevelopmental abnormalities in some patients |
| PASNA syndrome (OMIM no. 615474) | *CACNA1D* | Voltage-gated L-type calcium channel subunit $\alpha$-1D (Cav1.3) | Increased calcium influx | Early-onset PA, seizures, and neurological abnormalities; congenital hypoglycemia in some patients. |

ACTH indicates adrenocorticotropic hormone; ClC-2, chloride channel protein 2; FH, familial hyperaldosteronism; FH-I, familial hyperaldosteronism type I; FH-II, familial hyperaldosteronism type II; FH-III, familial hyperaldosteronism type III; FH-IV, familial hyperaldosteronism type IV; GIRK4, G-protein–activated inward rectifier potassium channel 4; GRA, glucocorticoid-remediable aldosteronism; and PA, primary aldosteronism.



and are in part shared between unilateral and bilateral forms of PA. Genetic risk loci have subsequently been replicated in a Japanese cohort and cross-ancestry meta-analysis,[18,96] together with additional risk loci, indicating a possible heterogeneity across different populations. Among them, Wnt2b is expressed in capsular cells of the adrenal cortex and its inactivation in mice results in a dysmorphic, hypocellular ZG and impaired aldosterone biosynthesis. This suggests that PA-associated risk loci predominantly disrupt adrenal cortex maintenance and cell differentiation.[97] Altogether, discovery of genetic risk loci represents a paradigm shift in our understanding of the genetics of PA, indicating that common genetic variation may predispose to aldosterone dysregulation in the general population, by modifying adrenocortical function and mineralocorticoid output in the adrenal gland. This could lead to a propitious environment for the appearance of somatic mutations and the development of PA in extreme cases.[17]

## DEMOGRAPHIC IMPLICATIONS FOR HYPERTENSION

Understanding the genetic causes of PA not only allows deciphering pathophysiological mechanisms but also allows addressing the diagnosis and management of hypertension in the general population. This is particularly relevant considering that PA is largely underdiagnosed, whereas it is early detection and targeted treatment allow potential cure of the disease in a significant proportion of patients and reduce cardiovascular morbidity overall.[5]

Genetic susceptibility and genetic mechanisms of PA may differ across different populations. Although direct comparison of the prevalence of PA among people with hypertension from different origins is unknown, it has been shown that relative aldosterone excess may contribute to early hypertension development in healthy, young Black adults in comparison to White people.[98] Genetic variants associated with PA may have different frequencies across different populations. This is particularly evident when considering the somatic genetics of APA, as described above, with *KCNJ5* mutations being more frequent in patients from Europe and Asia, while *CACNA1D* mutations predominate in patients from sub-Saharan African origin.[99–101] The relation of somatic mutations with age, sex, and hormonal status may have implications for the distribution and recognition of the disease across demographic strata, influencing both the apparent prevalence of PA and the likelihood of receiving an accurate diagnosis. Risk loci associated with PA are in part overlapping between European and Japanese cohorts, suggesting common molecular causes underlying increased susceptibility to developing the disease, whereas some differences have also been described.[17,18] This suggests that similar pathways may be effectively targeted in different populations for improved diagnosis and treatment, with space for specific approaches to reduce blood pressure and cardiovascular risk in patients with PA. However, different genetic susceptibility may contribute to variation in hypertension prevalence, severity, or the relative impact of aldosterone dysregulation across demographic groups, leading to health disparities because of unequal access to screening and treatment for PA. The genetic complexity of PA, with the coexistence of familial monogenic forms and sporadic disease with partly overlapping clinical presentation and age of onset, represents a diagnostic challenge, with FH being missed in the absence of targeted screening strategies. Finally,

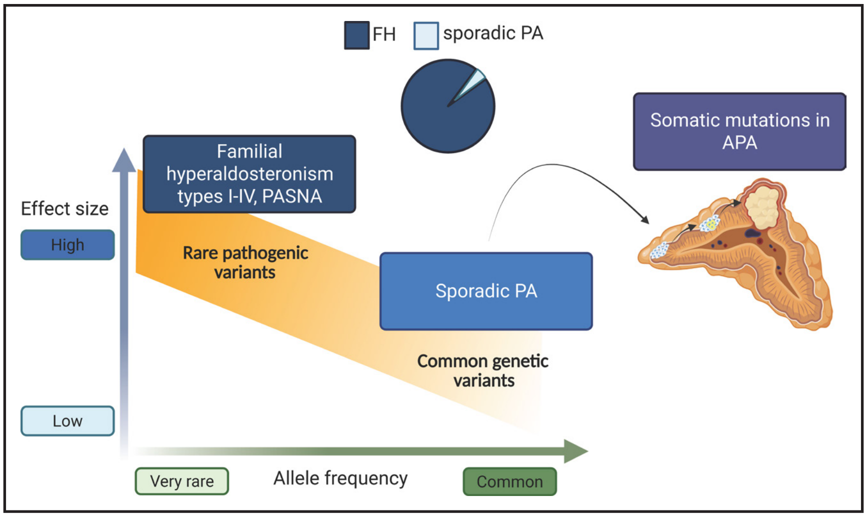


**Figure 2. The genetic spectrum of primary aldosteronism (PA) and aldosterone dysregulation.**
Rare genetic pathogenic variants with strong effect lead to Mendelian forms of PA, Familial hyperaldosteronism (FH) I to IV and PA, seizures, and neurological abnormalities (PASNA). On the opposite side of the spectrum, frequent alleles are associated with increased risk of developing PA in the general population, as discovered by genome-wide association studies, and may contribute to dysregulated aldosterone production by altering zona glomerulosa homeostasis. Finally, somatic mutations are found in a majority of aldosterone-producing adenoma (APA) and other aldosterone-producing structures in adrenals from patients with PA. The pathogenic link between genetic susceptibility and the development of APA remains still to be discovered.



genetic patterns may intersect with unequal access to screening and specialized care, raising the possibility that PA disproportionately contributes to resistant hypertension and cardiovascular risk in underserved populations.

## CONCLUSIONS AND OUTLOOK

PA has emerged as a highly prevalent and genetically heterogeneous cause of secondary hypertension, spanning a continuum from subtle aldosterone dysregulation in the general population to overt unilateral adenomas and familial forms (Figure 2). Advances in adrenal histopathology, steroidogenic biology, and high-resolution genomic approaches have revealed how age-related remodeling of adrenal zonation, the accumulation of somatic mutations across distinct adrenal structures, and the identification of genetic susceptibility factors converge on shared pathogenic mechanisms of aldosterone excess. These developments have refined our understanding of disease phenotypes and genotype–phenotype correlations and carry important implications for diagnosis, population-level screening, and therapeutic stratification. In particular, demographic trends and emerging susceptibility markers highlight a substantial reservoir of undiagnosed PA among individuals with hypertension. Continued integration of molecular, physiological, and epidemiological insights, together with large ongoing clinical studies investigating omics-based biomarkers of the disease,[102] is expected to improve diagnostic precision, support earlier and more individualized therapeutic strategies, and ultimately reduce the global burden of aldosterone-mediated cardiovascular disease.

## ARTICLE INFORMATION

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### Disclosures

None.

## REFERENCES


1. GBD. Risk Factors Collaborators. Global burden of 87 risk factors in 204 countries and territories, 1990-2019: a systematic analysis for the Global Burden of Disease Study 2019. *Lancet*. 2020;396:1223–1249. doi: 10.1016/S0140-6736(20)30752-2
2. NCD. Risk Factor Collaboration. Long-term and recent trends in hypertension awareness, treatment, and control in 12 high-income countries: an analysis of 123 nationally representative surveys. *Lancet*. 2019;394:639–651. doi: 10.1016/S0140-6736(19)31145-6
3. Zennaro MC, Boulkroun S, Fernandes-Rosa FL. Pathogenesis and treatment of primary aldosteronism. *Nat Rev Endocrinol*. 2020;16:578–589. doi: 10.1038/s41574-020-0382-4
4. Rossi GP, Rossitto G, Amar L, Azizi M, Riester A, Reincke M, Degenhart C, Widimsky J, Naruse M, Deinum J, et al. Drug-resistant hypertension in primary aldosteronism patients undergoing adrenal vein sampling: the AVIS-2-RH study. *Eur J Prev Cardiol*. 2022;29:e85–e93. doi: 10.1093/eurjpc/zwaa108
5. Adler GK, Stowasser M, Correa RR, Khan N, Kline G, McGowan MJ, Mulatero P, Murad MH, Touyz RM, Vaidya A, et al. Primary aldosteronism: an

endocrine society clinical practice guideline. *J Clin Endocrinol Metab*. 2025;110:2453–2495. doi: 10.1210/clinem/dgaf284
6. Monticone S, D'Ascenzo F, Moretti C, Williams TA, Veglio F, Gaita F, Mulatero P. Cardiovascular events and target organ damage in primary aldosteronism compared with essential hypertension: a systematic review and meta-analysis. *Lancet Diabetes Endocrinol*. 2018;6:41–50. doi: 10.1016/S2213-8587(17)30319-4
7. Freel EM, Mark PB, Weir RA, McQuarrie EP, Allan K, Dargie HJ, McClure JD, Jardine AG, Davies E, Connell JM. Demonstration of blood pressure-independent noninfarct myocardial fibrosis in primary aldosteronism: a cardiac magnetic resonance imaging study. *Circ Cardiovasc Imaging*. 2012;5:740–747. doi: 10.1161/CIRCIMAGING.112.974576
8. Davel AP, Jaffe IZ, Tostes RC, Jaisser F, Belin de Chantemele EJ. New roles of aldosterone and mineralocorticoid receptors in cardiovascular disease: translational and sex-specific effects. *Am J Physiol Heart Circ Physiol*. 2018;315:H989–H999. doi: 10.1152/ajpheart.00073.2018
9. Brown JM, Siddiqui M, Calhoun DA, Carey RM, Hopkins PN, Williams GH, Vaidya A. The unrecognized prevalence of primary aldosteronism: a cross-sectional study. *Ann Intern Med*. 2020;173:10–20. doi: 10.7326/M20-0065
10. Conn JW, Louis LH. Primary aldosteronism, a new clinical entity. *Ann Intern Med*. 1956;44:1–15. doi: 10.7326/0003-4819-44-1-1
11. Conn JW, Louis LH. Primary aldosteronism: a new clinical entity. *Trans Assoc Am Phys*. 1955;68:215–231, discussion, 231. doi:10.7326/0003-4819-44-1-1
12. Monticone S, Burrello J, Tizzani D, Bertello C, Viola A, Buffolo F, Gabetti L, Mengozzi G, Williams TA, Rabbia F, et al. Prevalence and clinical manifestations of primary aldosteronism encountered in primary care practice. *J Am Coll Cardiol*. 2017;69:1811–1820. doi: 10.1016/j.jacc.2017.01.052
13. de Freminville JB, Gardini M, Cremer A, Camelli S, Baron S, Bobrie G, Gosse P, Boulestreau R, Gebara N, Doublet J, et al. Prevalence and risk factors for secondary hypertension in young adults. *Hypertension*. 2024;81:2340–2349. doi: 10.1161/HYPERTENSIONAHA.124.22753
14. Fernandes-Rosa FL, Amar L, Tissier F, Bertherat J, Meatchi T, Zennaro MC, Boulkroun S. Functional histopathological markers of aldosterone producing adenoma and somatic KCNJ5 mutations. *Mol Cell Endocrinol*. 2015;408:220–226. doi: 10.1016/j.mce.2015.01.020
15. Yang J, Young MJ, Cole TJ, Fuller PJ. Mineralocorticoid receptor signalling in primary aldosteronism. *J Endocrinol*. 2023;259:e220249. doi: 10.1530/JOE-22-0249
16. Fernandes-Rosa FL, Boulkroun S, Fedlaoui B, Hureaux M, Travers-Allard S, Drossart T, Favier J, Zennaro MC. New advances in endocrine hypertension: from genes to biomarkers. *Kidney Int*. 2023;103:485–500. doi: 10.1016/j.kint.2022.12.021
17. Le Floch E, Cosentino T, Larsen CK, Beuschlein F, Reincke M, Amar L, Rossi GP, De Sousa K, Baron S, Chantalat S, et al. Identification of risk loci for primary aldosteronism in genome-wide association studies. *Nat Commun*. 2022;13:5198. doi: 10.1038/s41467-022-32896-8
18. Inoue K, Naito T, Fuji R, Sonehara K, Yamamoto K, Baba R, Kodama T, Otagaki Y, Okada A, Itcho K, et al. Primary aldosteronism and risk of cardiovascular outcomes: genome-wide association and Mendelian randomization study. *J Am Heart Assoc*. 2024;13:e034180. doi: 10.1161/JAHA.123.034180
19. Yates R, Katugampola H, Cavlan D, Cogger K, Meimaridou E, Hughes C, Metherell L, Guasti L, King P. Adrenocortical development, maintenance, and disease. *Curr Top Dev Biol*. 2013;106:239–312. doi: 10.1016/B978-0-12-416021-7.00007-9
20. Abou Nader N, Boyer A. Adrenal cortex development and maintenance: knowledge acquired from mouse models. *Endocrinology*. 2021;162:bqab187. doi: 10.1210/endocr/bqab187
21. Kim JH, Choi MH. Embryonic development and adult regeneration of the adrenal gland. *Endocrinol Metab (Seoul)*. 2020;35:765–773. doi: 10.3803/EnM.2020.403
22. Beuschlein F, Mutch C, Bavers DL, Ulrich-Lai YM, Engeland WC, Keegan C, Hammer GD. Steroidogenic factor-1 is essential for compensatory adrenal growth following unilateral adrenalectomy. *Endocrinology*. 2002;143:3122–3135. doi: 10.1210/endo.143.8.8944
23. Luo X, Ikeda Y, Parker KL. A cell-specific nuclear receptor is essential for adrenal and gonadal development and sexual differentiation. *Cell*. 1994;77:481–490. doi: 10.1016/0092-8674(94)90211-9
24. Freedman BD, Kempna PB, Carlone DL, Shah M, Guagliardo NA, Barrett PQ, Gomez-Sanchez CE, Majzoub JA, Breault DT. Adrenocortical zonation results from lineage conversion of differentiated zona glomerulosa cells. *Dev Cell*. 2013;26:666–673. doi: 10.1016/j.devcel.2013.07.016
25. Schimmer BP, White PC. Minireview: steroidogenic factor 1: its roles in differentiation, development, and disease. *Mol Endocrinol*. 2010;24:1322–1337. doi: 10.1210/me.2009-0519
26. Dumontet T, Sahut-Barnola I, Septier A, Montanier N, Plotton I, Roucher-Boulez F, Ducros V, Lefrançois-Martinez AM, Pointud JC, Zubair M, et al. PKA signaling drives reticularis differentiation and sexually dimorphic adrenal cortex renewal. *JCI Insight*. 2018;3:e98394. doi: 10.1172/jci.insight.98394
27. Lyraki R, Schedl A. Adrenal cortex renewal in health and disease. *Nat Rev Endocrinol*. 2021;17:421–434. doi: 10.1038/s41574-021-00491-4
28. Miller WL, Auchus RJ. The molecular biology, biochemistry, and physiology of human steroidogenesis and its disorders. *Endocr Rev*. 2011;32:81–151. doi: 10.1210/er.2010-0013
29. Connell JMC, Davies E. The new biology of aldosterone. *J Endocrinol*. 2005;186:1–20. doi: 10.1677/joe.1.06017
30. Brown NJ. Contribution of aldosterone to cardiovascular and renal inflammation and fibrosis. *Nat Rev Nephrol*. 2013;9:459–469. doi: 10.1038/nrneph.2013.110
31. Spät A, Hunyady L. Control of aldosterone secretion: a model for convergence in cellular signaling pathways. *Physiol Rev*. 2004;84:489–539. doi: 10.1152/physrev.00030.2003
32. Ruggiero C, Lalli E. Impact of ACTH signaling on transcriptional regulation of steroidogenic genes. *Front Endocrinol*. 2016;7:24. doi: 10.3389/fendo.2016.00024
33. Stathori G, Alexakis D, Chrousos GP, Paltoglou G. Examining aldosterone plasma concentration alterations post-ACTH stimulation in healthy subjects: a systematic literature review and meta-analysis on ACTH's role in aldosterone secretion. *Hormones (Athens)*. 2024;23:765–775. doi: 10.1007/s42000-024-00583-6
34. Fuchs-Hammoser R, Schweiger M, Oelkers W. The effect of chronic low-dose infusion of ACTH (1-24) on renin, renin-substrate, aldosterone and other corticosteroids in sodium replete and deplete man. *Acta Endocrinologica*. 1980;95:198–206. doi: 10.1530/acta.0.0950198
35. Boulkroun S, Samson-Couterie B, Dzib JF, Lefebvre H, Louiset E, Amar L, Plouin PF, Lalli E, Jeunemaitre X, Benecke A, et al. Adrenal cortex remodeling and functional zona glomerulosa hyperplasia in primary aldosteronism. *Hypertension*. 2010;56:885–892. doi: 10.1161/HYPERTENSIONAHA.110.158543
36. Nishimoto K, Nakagawa K, Li D, Kosaka T, Oya M, Mikami S, Shibata H, Itoh H, Mitani F, Yamazaki T, et al. Adrenocortical zonation in humans under normal and pathological conditions. *J Clin Endocrinol Metab*. 2010;95:2296–2305. doi: 10.1210/jc.2009-2010
37. Williams TA, Gomez-Sanchez CE, Rainey WE, Giordano TJ, Lam AK, Marker A, Mete O, Yamazaki Y, Zerbini MCN, Beuschlein F, et al. International histopathology consensus for unilateral primary aldosteronism. *J Clin Endocrinol Metab*. 2021;106:42–54. doi: 10.1210/clinem/dgaa484
38. Boulkroun S, Samson-Couterie B, Golib-Dzib JF, Amar L, Plouin PF, Sibony M, Lefebvre H, Louiset E, Jeunemaitre X, Meatchi T, et al. Aldosterone-producing adenoma formation in the adrenal cortex involves expression of stem/progenitor cell markers. *Endocrinology*. 2011;152:4753–4763. doi: 10.1210/en.2011-1205
39. Nishimoto K, Tomlins SA, Kuick R, Cani AK, Giordano TJ, Hovelson DH, Liu CJ, Sanjanwala AR, Edwards MA, Gomez-Sanchez CE, et al. Aldosterone-stimulating somatic gene mutations are common in normal adrenal glands. *Proc Natl Acad Sci USA*. 2015;112:E4591–E4599. doi: 10.1073/pnas.1505529112
40. Omata K, Anand SK, Hovelson DH, Liu CJ, Yamazaki Y, Nakamura Y, Ito S, Satoh F, Sasano H, Rainey WE, et al. Aldosterone-producing cell clusters frequently harbor somatic mutations and accumulate with age in normal adrenals. *J Endocr Soc*. 2017;1:787–799. doi: 10.1210/js.2017-00134
41. Nanba K, Vaidya A, Williams GH, Zheng I, Else T, Rainey WE. Age-related autonomous aldosteronism. *Circulation*. 2017;136:347–355. doi: 10.1161/CIRCULATIONAHA.117.028201
42. Spyroglou A, Piaditis GP, Kaltsas G, Alexandraki KI. Transcriptomics, epigenetics, and metabolomics of primary aldosteronism. *Cancers*. 2021;13:5582. doi: 10.3390/cancers13215582
43. Murakami M, Rhayem Y, Kunzke T, Sun N, Feuchtinger A, Ludwig P, Strom TM, Gomez-Sanchez C, Knosel T, Kirchner T, et al. In situ metabolomics of aldosterone-producing adenomas. *JCI Insight*. 2019;4:e130356. doi: 10.1172/jci.insight.130356
44. Yoshii Y, Oki K, Gomez-Sanchez CE, Ohno H, Itcho K, Kobuke K, Yoneda M. Hypomethylation of CYP11B2 in aldosterone-producing adenoma. *Hypertension*. 2016;68:1432–1437. doi: 10.1161/HYPERTENSIONAHA.116.08313
45. Rege J, Bandulik S, Nanba K, Kosmann C, Blinder AR, Plain A, Vats P, Kumar-Sinha C, Lerario AM, Else T, et al. Somatic SLC30A1 mutations altering zinc transporter ZnT1 cause aldosterone-producing adenomas and primary aldosteronism. *Nat Genet*. 2023;55:1623–1631. doi: 10.1038/s41588-023-01498-5
46. van Rooyen D, Bandulik S, Coon GA, Laukemper M, Kumar-Sinha C, Udager AM, Lerario AM, Lee C, Wachtel H, Cohen DL, et al. Somatic mutations in MCOLN3

are associated with aldosterone-producing adenomas. *Hypertension*. 2025;82:1778–1788. doi: 10.1161/HYPERTENSIONAHA.125.24909
47. Zhou J, Azizan EAB, Cabrera CP, Fernandes-Rosa FL, Boulkroun S, Argentesi G, Cottrell E, Amar L, Wu X, O'Toole S, et al. Somatic mutations of GNA11 and GNAQ in CTNNB1-mutant aldosterone-producing adenomas presenting in puberty, pregnancy or menopause. *Nat Genet*. 2021;53:1360–1372. doi: 10.1038/s41588-021-00906-y
48. Wu X, Azizan EAB, Goodchild E, Garg S, Hagiyama M, Cabrera CP, Fernandes-Rosa FL, Boulkroun S, Kuan JL, Tiang Z, et al. Somatic mutations of CADM1 in aldosterone-producing adenomas and gap junction-dependent regulation of aldosterone production. *Nat Genet*. 2023;55:1009–1021. doi: 10.1038/s41588-023-01403-0
49. Fernandes-Rosa FL, Giscos-Douriez I, Amar L, Gomez-Sanchez CE, Meatchi T, Boulkroun S, Zennaro MC. Different somatic mutations in multinodular adrenals with aldosterone-producing adenoma. *Hypertension*. 2015;66:1014–1022. doi: 10.1161/HYPERTENSIONAHA.115.05993
50. De Sousa K, Boulkroun S, Baron S, Nanba K, Wack M, Rainey WE, Rocha A, Giscos-Douriez I, Meatchi T, Amar L, et al. Genetic, cellular, and molecular heterogeneity in adrenals with aldosterone-producing adenoma. *Hypertension*. 2020;75:1034–1044. doi: 10.1161/HYPERTENSIONAHA.119.14177
51. De Sousa K, Abdellatif AB, Giscos-Douriez I, Meatchi T, Amar L, Fernandes-Rosa FL, Boulkroun S, Zennaro MC. Colocalization of Wnt/beta-catenin and ACTH signaling pathways and paracrine regulation in aldosterone-producing adenoma. *J Clin Endocrinol Metab*. 2022;107:419–434. doi: 10.1210/clinem/dgab707
52. Sun Z, Jing C, Tetti M, Gong S, Wei J, Pang Y, Reincke M, Williams TA. Integrated transcriptomics reveals evolutionary trajectories and cell density-dependent mechanisms in aldosterone-producing adenomas. *Adv Sci*. 2026;e05410. doi: 10.1002/advs.202505410
53. Omata K, Satoh F, Morimoto R, Ito S, Yamazaki Y, Nakamura Y, Anand SK, Guo Z, Stowasser M, Sasano H, et al. Cellular and genetic causes of idiopathic hyperaldosteronism. *Hypertension*. 2018;72:874–880. doi: 10.1161/HYPERTENSIONAHA.118.11086
54. Choi M, Scholl UI, Yue P, Björklund P, Zhao B, Nelson-Williams C, Ji W, Cho Y, Patel A, Men CJ, et al. K+ channel mutations in adrenal aldosterone-producing adenomas and hereditary hypertension. *Science*. 2011;331:768–772. doi: 10.1126/science.1198785
55. Scholl UI, Stölting G, Schewe J, Thiel A, Tan H, Nelson-Williams C, Vichot AA, Jin SC, Loring E, Untiet V, et al. CLCN2 chloride channel mutations in familial hyperaldosteronism type II. *Nat Genet*. 2018;50:349–354. doi: 10.1038/s41588-018-0048-5
56. Fernandes-Rosa FL, Daniil G, Orozco IJ, Göppner C, El Zein R, Jain V, Boulkroun S, Jeunemaitre X, Amar L, Lefebvre H, et al. A gain-of-function mutation in the CLCN2 chloride channel gene causes primary aldosteronism. *Nat Genet*. 2018;50:355–361. doi: 10.1038/s41588-018-0053-8
57. Göppner C, Orozco IJ, Hoegg-Beiler MB, Soria AH, Hübner CA, Fernandes-Rosa FL, Boulkroun S, Zennaro MC, Jentsch TJ. Pathogenesis of hypertension in a mouse model for human CLCN2 related hyperaldosteronism. *Nat Commun*. 2019;10:4678. doi: 10.1038/s41467-019-12113-9
58. Stindl J, Tauber P, Sterner C, Tegtmeier I, Warth R, Bandulik S. Pathogenesis of adrenal aldosterone-producing adenomas carrying mutations of the Na(+)/K(+)-ATPase. *Endocrinology*. 2015;156:4582–4591. doi: 10.1210/en.2015-1466
59. Azizan EAB, Poulsen H, Tuluc P, Zhou J, Clausen MV, Lieb A, Maniero C, Garg S, Bochukova EG, Zhao W, et al. Somatic mutations in ATP1A1 and CACNA1D underlie a common subtype of adrenal hypertension. *Nat Genet*. 2013;45:1055–1060. doi: 10.1038/ng.2716
60. Scholl UI, Goh G, Stölting G, de Oliveira RC, Choi M, Overton JD, Fonseca AL, Korah R, Starker LF, Kunstman JW, et al. Somatic and germline CACNA1D calcium channel mutations in aldosterone-producing adenomas and primary aldosteronism. *Nat Genet*. 2013;45:1050–1054. doi: 10.1038/ng.2695
61. Scholl UI, Stölting G, Nelson-Williams C, Vichot AA, Choi M, Loring E, Prasad ML, Goh G, Carling T, Juhlin CC, et al. Recurrent gain of function mutation in calcium channel CACNA1H causes early-onset hypertension with primary aldosteronism. *eLife*. 2015;4:e06315. doi: 10.7554/eLife.06315
62. Daniil G, Fernandes-Rosa FL, Chemin J, Blesneac I, Beltrand J, Polak M, Jeunemaitre X, Boulkroun S, Amar L, Strom TM, et al. CACNA1H mutations are associated with different forms of primary aldosteronism. *EBioMedicine*. 2016;13:225–236. doi: 10.1016/j.ebiom.2016.10.002
63. Williams TA, Monticone S, Schack VR, Stindl J, Burrello J, Buffolo F, Annaratone L, Castellano I, Beuschlein F, Reincke M, et al. Somatic ATP1A1, ATP2B3, and KCNJ5 mutations in aldosterone-producing adenomas. *Hypertension*. 2014;63:188–195. doi: 10.1161/HYPERTENSIONAHA.113.01733
64. Berthon A, Drelon C, Ragazzon B, Boulkroun S, Tissier F, Amar L, Samson-Couterie B, Zennaro MC, Plouin PF, Skah S, et al. WNT/β-catenin signalling is activated in aldosterone-producing adenomas and controls aldosterone production. *Hum Mol Genet*. 2014;23:889–905. doi: 10.1093/hmg/ddt484
65. St-Jean M, Bourdeau I, Martin M, Lacroix A. Aldosterone is aberrantly regulated by various stimuli in a high proportion of patients with primary aldosteronism. *J Clin Endocrinol Metab*. 2021;106:e45–e60. doi: 10.1210/clinem/dgaa703
66. Boulkroun S, Beuschlein F, Rossi GP, Golib-Dzib JF, Fischer E, Amar L, Mulatero P, Samson-Couterie B, Hahner S, Quinkler M, et al. Prevalence, clinical, and molecular correlates of KCNJ5 mutations in primary aldosteronism. *Hypertension*. 2012;59:592–598. doi: 10.1161/HYPERTENSIONAHA.111.186478
67. Fernandes-Rosa FL, Williams TA, Riester A, Steichen O, Beuschlein F, Boulkroun S, Strom TM, Monticone S, Amar L, Meatchi T, et al. Genetic spectrum and clinical correlates of somatic mutations in aldosterone-producing adenoma. *Hypertension*. 2014;64:354–361. doi: 10.1161/HYPERTENSIONAHA.114.03419
68. Lenzini L, Rossitto G, Maiolino G, Letizia C, Funder JW, Rossi GP. A meta-analysis of somatic KCNJ5 $K^+$ channel mutations in 1636 patients with an aldosterone-producing adenoma. *J Clin Endocrinol Metab*. 2015;100:E1089–E1095. doi: 10.1210/jc.2015-2149
69. Chang YY, Tsai CH, Peng SY, Chen ZW, Chang CC, Lee BC, Liao CW, Pan CT, Chen YL, Lin LC, et al; TAIPAI Study Group. KCNJ5 somatic mutations in aldosterone-producing adenoma are associated with a worse baseline status and better recovery of left ventricular remodeling and diastolic function. *Hypertension*. 2021;77:114–125. doi: 10.1161/hypertensionaha.120.15679
70. Vilela LAP, Rassi-Cruz M, Guimaraes AG, Moises CCS, Freitas TC, Alencar NP, Petenuci J, Goldbaum TS, Maciel AAW, Pereira MAA, et al. KCNJ5 somatic mutation is a predictor of hypertension remission after adrenalectomy for unilateral primary aldosteronism. *J Clin Endocrinol Metab*. 2019;104:4695–4702. doi: 10.1210/jc.2019-00531
71. Nanba K, Baker JE, Blinder AR, Bick NR, Liu CJ, Lim JS, Wachtel H, Cohen DL, Williams TA, Reincke M, et al. Histopathology and genetic causes of primary aldosteronism in young adults. *J Clin Endocrinol Metab*. 2022;107:2473–2482. doi: 10.1210/clinem/dgac408
72. Blinder AR, Sapiro D, Baker JE, Qin Z, Bancos I, Lyden ML, Young WF, Wachtel H, Cohen DL, Kürzinger L, et al. Adrenal histologic and genetic characteristics of primary aldosteronism in late adulthood. *Eur J Endocrinol*. 2026;194:208–216. doi: 10.1093/ejendo/lvag022
73. Lopez A, Duparc C, Renouf S, D'Agostino M, De Sousa K, Amar L, Defortescu G, Manceau G, Sabourin J, Fernandes-Rosa FL, et al. Regulation of aldosterone secretion by substance P and the neurokinin type 1 receptor in aldosterone-producing adenomas. *J Am Heart Assoc*. 5539;15:e04. doi: 10.1161/jaha.125.045539
74. Gong S, Sun N, Meyer LS, Tetti M, Koupourtidou C, Krebs S, Masserdotti G, Blum H, Rainey WE, Reincke M, et al. Primary aldosteronism: spatial multiomics mapping of genotype-dependent heterogeneity and tumor expansion of aldosterone-producing adenomas. *Hypertension*. 2023;80:1555–1567. doi: 10.1161/HYPERTENSIONAHA.123.20921
75. Murakami M, Hara K, Ikeda K, Horino M, Okazaki R, Niitsu Y, Takeuchi A, Aoki J, Shiba K, Tsujimoto K, et al. Single-nucleus analysis reveals tumor heterogeneity of aldosterone-producing adenoma. *Hypertension*. 2024;81:361–371. doi: 10.1161/HYPERTENSIONAHA.123.21446
76. Yokomoto-Umakoshi M, Fujita M, Umakoshi H, Ogasawara T, Iwahashi N, Nakatani K, Kaneko H, Fukumoto T, Nakao H, Haji S, et al. Multiomics analysis unveils the cellular ecosystem with clinical relevance in aldosterone-producing adenomas with KCNJ5 mutations. *Proc Natl Acad Sci USA*. 2025;122:e2421489122. doi: 10.1073/pnas.2421489122
77. Mulatero P, Scholl UI, Fardella CE, Charmandari E, Januszewicz A, Reincke M, Gomez-Sanchez CE, Stowasser M, Dekkers OM. Familial hyperaldosteronism: an European reference network on rare endocrine conditions clinical practice guideline. *Eur J Endocrinol*. 2024;190:G1–G14. doi: 10.1093/ejendo/lvae041
78. Lifton RP, Dluhy RG, Powers M, Rich GM, Cook S, Ulick S, Lalouel JM. A chimaeric 11beta-hydroxylase aldosterone synthase gene causes glucocorticoid-remediable aldosteronism and human hypertension. *Nature*. 1992;355:262–265. doi: 10.1038/355262a0
79. Pascoe L, Curnow KM, Slutsker L, Connell JM, Speiser PW, New MI, White PC. Glucocorticoid-suppressible hyperaldosteronism results from hybrid genes created by unequal crossovers between CYP11B1 and CYP11B2. *Proc Natl Acad Sci USA*. 1992;89:8327–8331. doi: 10.1073/pnas.89.17.8327
80. Stowasser M, Bachmann AW, Huggard PR, Rossetti TR, Gordon RD. Treatment of familial hyperaldosteronism type I: only partial suppression of adrenocorticotropin required to correct hypertension. *J Clin Endocrinol Metab*. 2000;85:3313–3318. doi: 10.1210/jcem.85.9.6834

81. Sutherland DJ, Ruse JL, Laidlaw JC. Hypertension, increased aldosterone secretion and low plasma renin activity relieved by dexamethasone. *Can Med Assoc J*. 1966;95:1109–1119.
82. Gordon RD, Stowasser M, Tunny TJ, Klemm SA, Finn WL, Krek AL. Clinical and pathological diversity of primary aldosteronism, including a new familial variety. *Clin Exp Pharmacol Physiol*. 1991;18:283–286. doi: 10.1111/j.1440-1681.1991.tb01446.x
83. Stowasser M, Gordon RD, Tunny TJ, Klemm SA, Finn WL, Krek AL. Familial hyperaldosteronism type II: five families with a new variety of primary aldosteronism. *Clin Exp Pharmacol Physiol*. 1992;19:319–322. doi: 10.1111/j.1440-1681.1992.tb00462.x
84. Dutta RK, Arnesen T, Heie A, Walz M, Alesina P, Söderkvist P, Gimm O. A somatic mutation in CLCN2 identified in a sporadic aldosterone-producing adenoma. *Eur J Endocrinol*. 2019;181:K37–K41. doi: 10.1530/EJE-19-0377
85. Mulatero P, Monticone S, Rainey WE, Veglio F, Williams TA. Role of KCNJ5 in familial and sporadic primary aldosteronism. *Nat Rev Endocrinol*. 2013;9:104–112. doi: 10.1038/nrendo.2012.230
86. Monticone S, Tetti M, Burrello J, Buffolo F, De Giovanni R, Veglio F, Williams TA, Mulatero P. Familial hyperaldosteronism type III. *J Hum Hypertens*. 2017;31:776–781. doi: 10.1038/jhh.2017.34
87. Azizan EAB, Drake WM, Brown MJ. Primary aldosteronism: molecular medicine meets public health. *Nat Rev Nephrol*. 2023;19:788–806. doi: 10.1038/s41581-023-00753-6
88. Mulatero P, Tauber P, Zennaro MC, Monticone S, Lang K, Beuschlein F, Fischer E, Tizzani D, Pallauf A, Viola A, et al. KCNJ5 mutations in European families with nonglucocorticoid remediable familial hyperaldosteronism. *Hypertension*. 2012;59:235–240. doi: 10.1161/HYPERTENSIONAHA.111.183996
89. Scholl UI, Nelson-Williams C, Yue P, Grekin R, Wyatt RJ, Dillon MJ, Couch R, Hammer LK, Harley FL, Farhi A, et al. Hypertension with or without adrenal hyperplasia due to different inherited mutations in the potassium channel KCNJ5. *Proc Natl Acad Sci USA*. 2012;109:2533–2538. doi: 10.1073/pnas.1121407109
90. Monticone S, Hattangady NG, Penton D, Isales CM, Edwards MA, Williams TA, Sterner C, Warth R, Mulatero P, Rainey WE. A novel Y152C KCNJ5 mutation responsible for familial hyperaldosteronism type III. *J Clin Endocrinol Metab*. 2013;98:E1861–E1865. doi: 10.1210/jc.2013-2428
91. De Mingo Alemany MC, Mifsud Grau L, Moreno Macián F, Ferrer Lorente B, León Cariñena S. A de novo CACNA1D missense mutation in a patient with congenital hyperinsulinism, primary hyperaldosteronism and hypotonia. *Channels (Austin)*. 2020;14:175–180. doi: 10.1080/19336950.2020.1761171
92. Török F, Tezcan K, Filippini L, Fernández-Quintero ML, Zanetti L, Liedl KR, Drexel RS, Striessnig J, Ortner NJ. Germline de novo variant F747S extends the phenotypic spectrum of CACNA1D $Ca^{2+}$ channelopathies. *Hum Mol Genet*. 2023;32:847–859. doi: 10.1093/hmg/ddac248
93. Brown JM, Honzel B, Tsai LC, Milks J, Neibuhr YM, Newman AJ, Cherney M, Stouffer DG, Auchus RJ, Vaidya A. Characterizing the origins of primary aldosteronism. *Hypertension*. 2025;82:306–318. doi: 10.1161/HYPERTENSIONAHA.124.24153
94. Williams TA, Lenders JWM, Mulatero P, Burrello J, Rottenkolber M, Adolf C, Satoh F, Amar L, Quinkler M, Deinum J, et al; Primary Aldosteronism Surgery Outcome (PASO) Investigators. Outcomes after adrenalectomy for unilateral primary aldosteronism: an international consensus on outcome measures and analysis of remission rates in an international cohort. *Lancet Diabetes Endocrinol*. 2017;5:689–699. doi: 10.1016/S2213-8587(17)30135-3
95. Hacini I, De Sousa K, Boulkroun S, Meatchi T, Amar L, Zennaro MC, Fernandes-Rosa FL. Somatic mutations in adrenals from patients with primary aldosteronism not cured after adrenalectomy suggest common pathogenic mechanisms between unilateral and bilateral disease. *Eur J Endocrinol*. 2021;185:405–412. doi: 10.1530/EJE-21-0338
96. Naito T, Inoue K, Sonehara K, Baba R, Kodama T, Otagaki Y, Okada A, Itcho K, Kobuke K, Kishimoto S, et al. Genetic risk of primary aldosteronism and its contribution to hypertension: a cross-ancestry meta-analysis of genome-wide association studies. *Circulation*. 2023;147:1097–1109. doi: 10.1161/CIRCULATIONAHA.122.062349
97. Borges KS, Little DW, Magalhães T de A, Ribeiro C, Dumontet T, Lapensee C, Basham KJ, Seth A, Azova S, Guagliardo NA, et al. Non-canonical Wnt signaling triggered by WNT2B drives adrenal aldosterone production. *bioRxiv*. Preprint posted online August 24, 2024. doi: 10.1101/2024.08.23.609423
98. Gafane-Matemane LF, Mokae NL, Breet Y, Poglitsch M, Schutte AE. Associations of central and peripheral blood pressure with the renin-angiotensin-aldosterone system in healthy young adults: the African-PREDICT study. *Hypertens Res*. 2021;44:435–445. doi: 10.1038/s41440-020-00566-1
99. Nanba K, Yamazaki Y, Bick N, Onodera K, Tezuka Y, Omata K, Ono Y, Blinder AR, Tomlins SA, Rainey WE, et al. Prevalence of somatic mutations in aldosterone-producing adenomas in Japanese patients. *J Clin Endocrinol Metab*. 2020;105:e4066–e4073. doi: 10.1210/clinem/dgaa595
100. Nanba K, Omata K, Else T, Beck PCC, Nanba AT, Turcu AF, Miller BS, Giordano TJ, Tomlins SA, Rainey WE. Targeted molecular characterization of aldosterone-producing adenomas in White Americans. *J Clin Endocrinol Metab*. 2018;103:3869–3876. doi: 10.1210/jc.2018-01004
101. Nanba K, Omata K, Gomez-Sanchez CE, Stratakis CA, Demidowich AP, Suzuki M, Thompson LDR, Cohen DL, Luther JM, Gellert L, et al. Genetic characteristics of aldosterone-producing adenomas in Blacks. *Hypertension*. 2019;73:885–892. doi: 10.1161/HYPERTENSIONAHA.118.12070
102. Reel PS, Reel S, van Kralingen JC, Langton K, Lang K, Erlic Z, Larsen CK, Amar L, Pamporaki C, Mulatero P, et al. Machine learning for classification of hypertension subtypes using multi-omics: a multi-centre, retrospective, data-driven study. *EBioMedicine*. 2022;84:104276. doi: 10.1016/j.ebiom.2022.104276